

SHIH-NI PRIM

sprim@ncsu.edu
(336) 254-3431

EDUCATION

PhD	North Carolina State University, Statistics Advisor: Brian Reich Cumulative GPA: 4.0	2022-present
MR	North Carolina State University, Statistics Cumulative GPA: 4.0	Dec 2022

RESEARCH INTERESTS

Bayesian inference and hierarchical modeling, spatial and spatiotemporal statistics, tensor decomposition, tensor regression, surrogate modeling, active learning, sequential design, climate and environmental applications, causal inference, probabilistic machine learning

CURRENT RESEARCH PROJECTS

With Brian Reich and Yawen Guan, “A Multivariate Spectral Model with Adjustment for Unmeasured Confounders”: To find causal effects, we propose a spectral method that uses Bayesian hierarchical models to adjust for spatial confounding. We use tensor coefficients to allow for them to vary by frequency and shrinkage prior for variable selection. We intend to analyze the effects of environmental and socioeconomic factors on health outcomes.

With Kevin Quinlan and Annie Booth. “Active Learning with Deep Gaussian Processes for Aero Database Construction”: We explore whether deep Gaussian Processes (GP) perform better than ordinary GP as surrogate models for aerodynamic data. We also devise new algorithms for active learning to more efficiently construct aero databases.

AWARDS

B. B. Bhattacharyya Excellence Award North Carolina State University, Department of Statistics	2023–2024
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COMPUTER SKILLS

Programming: R, Python, SAS, Matlab, Java
Certificate: SAS Certified Specialist: Base Programming Using SAS 9.4 (Issued May 2020)

RESEARCH EXPERIENCE

Environmental Protection Agency , Chapel Hill, NC	August 2024 to present
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Oak Ridge Institute for Science and Education (ORISE) Fellow

- Develop statistical models to characterize effects of PFAS on multiple health outcomes
- Participate in meetings and collaborate with team members

Lawrence Livermore National Laboratory, Livermore, CA
Data Science Summer Institute Intern

May to August 2024

North Carolina State University, Raleigh, NC

August 2022 to May 2024

Research Trainee at Duke Clinical Research Institute

- Helped with developing and evaluating analytic methods based on WIN statistics
- Shadowed meetings of clinical research and received training on ethics of research

North Carolina State University, Raleigh, NC
Research Assistant

Summer 2022, July 2023 to May 2024

- Principal Investigators: Natalie Nelson (Department of Biological and Agricultural Engineering) and Brian Reich (Department of Statistics)
- Used statistical/machine learning models to analyze data about harmful algal blooms
- Data cleaning, data analysis, statistical modeling, manuscript drafting

Wake Forest Baptist Health, Winston-Salem, NC

2021 to 2022

Volunteer Research Statistician / Intern

- Department of Radiation Oncology
 - Conduct Kaplan Meier Survival Analysis and Cox Proportional Hazard Model on retrospective data of patients with glioblastoma
 - Create tables and plots and help with writing and editing for publication
- Radiology Informatics and Image Processing Laboratory
 - Help collect MRI images and clinical data for studies that utilize machine learning for classification

PUBLICATIONS

Chan MD, **Prim S**, Cramer C, Ruiz J. “Nonrandomized trials in clinical oncology.” In *Handbook for Designing and Conducting Clinical and Translational Research: Translational Radiation Oncology*, edited by Adam E. M. Eltorai, Jeffrey A. Bakal, Daniel W. Kim, David E. Wazer, 313–284. 2023. Academic Press.

Helis CA, **Prim S**, Cramer CK, Strowd R, Lesser GJ, White JJ, Tatter SB, Laxton AW, Whitlow C, Lo H, Debinski W, Ververs JD, Black PJ, Chan MD, “Clinical outcomes of dose-escalated re-irradiation in patients with recurrent high-grade glioma.” *Neuro-Oncology Practice* (2022). <https://doi.org/10.1093/nop/npac032>.

Severino C, Egan D, **Prim S**. “International Undergraduates’ Perceptions of their Second Language Writing Development and Their Implications for Writing Center Tutors.” *Writing Center Journal* 38/1-2 (2020): 165–202.

Prim S. “The Utah Symphony’s Recordings with Vanguard Records.” *Association for Recorded Sound Collections Journal* 48/1 (2017): 23–41.

Severino C, **Prim S.** “Second Language Writing Development and the Role of Tutors: A Case Study of an Online Writing Center ‘Frequent Flyer.’” *Writing Center Journal* 35, no. 3 (2016): 143–185.

Severino C, **Prim S.** “Word Choice Errors in Chinese Students’ English Writing and How Online Writing Center Tutors Respond to Them.” *Writing Center Journal* 34, no. 2 (2015): 115–143.

CONFERENCE PRESENTATIONS

Prim S., Quinlan K, Booth A. “Active Learning with Deep Gaussian Processes for Aero Database Construction.” International Conference on Advances in Interdisciplinary Statistics and Combinatorics, October 13, 2024.

Severino C, Egan D, **Prim S.** “International Undergraduates’ Perceptions of their Second Language Writing Development.” International Writing Centers Association, Chicago, November 12, 2017.

Prim S. “The Utah Symphony’s 1971 Goodwill Tour in Latin America: Cultural Diplomacy during the Cold War.” American Musicological Society Southeast Chapter Spring Meeting, Furman University, Greenville, SC, March 18, 2017.

Severino C, **Prim S.** “Second Language Writing Development and the Role of Tutors: A Case Study of an Online Writing Center ‘Frequent Flyer.’” Midwest Writing Centers Associations, Cedar Rapids, IA, March 4, 2016.

Prim S. “Maurice Abravanel and Gustav Mahler: The Reception of Early Mahler Recordings by Abravanel and the Utah Symphony Orchestra.” American Musicological Society Southeast Chapter Spring Meeting, University of North Carolina at Chapel Hill, NC, February 23, 2014.

PROFESSIONAL SERVICE

Book Reviewer
Reviewer
Grant Reviewer

Notes (2018 to 2021)
Writing Center Journal (2015 to 2016)
University of Iowa (2013)

GRANTS AND FELLOWSHIPS

ORISE Fellowship
Environmental Protection Agency

2024 to present

T32 Training Grant Duke Clinical Research Institute	2022 to 2024
Travel Grant Midwest Writing Centers Association	2016
Travel Grant Executive Council of Graduate & Professional Students, University of Iowa	2013
Study Abroad Award Ministry of Education, Taiwan	2010 to 2012

LANGUAGES

English, Mandarin, German, French, Italian

REFERENCES

Brian Reich, Gertrude M Cox Distinguished Professor of Statistics
Department of Statistics, North Carolina State University
Phone: 919-513-7686
Email: bjreich@ncsu.edu

Yawen Guan, Assistant Professor
Department of Statistics, Colorado State University
Email: yawen.guan@colostate.edu

Natalie Nelson, Associate Professor
Department of Biological and Agricultural Engineering, North Carolina State University
Phone: 919-515-6741
Email: nnelson4@ncsu.edu

OTHER EDUCATION

PhD	University of Iowa, Musicology Dissertation: “Maurice Abravanel and the Utah Symphony’s Performances and Recordings of Gustav Mahler’s Symphonies (1951–1979)” (Advisor: Nathan Platte)	May 2016
MM	Florida State University, Musicology Thesis: “Gustav Mahler’s <i>Das Lied von der Erde</i> : An Intellectual Journey across Cultures and Beyond Life and Death” (Advisor: Douglass Seaton)	May 2009
MM	University of North Carolina at Greensboro, Saxophone Performance	May 2006
MA	University of North Carolina at Chapel Hill, Comparative Literature Thesis: “Music-Myth into Comedy: How Shaw Used Wagner to Transform Mozart’s Legend” (Advisor: William Harmon)	May 2004

TEACHING EXPERIENCE

North Carolina State University, Raleigh, NC

May to August 2022

Teaching Assistant, Department of Statistics

- Assist with two graduate courses: Fundamentals of Statistical Inference I and Fundamentals of Linear Models and Regression
- Duties: Grade homework and hold weekly office hours

Wake Forest University, Winston-Salem, NC

August 2017 to December 2017

Adjunct Faculty, Department of Music

- Independently designed and taught the course “Introduction to Western Music,” which covered western music history from the Middle Ages to the Twentieth Century
- Used music as artifacts to help students think about history in the appropriate contexts and cultivate critical thinking
- Provided necessary study aids including detailed guidelines and rubrics for writing assignments and study guides for exams
- Used a scaffolding method to help students write research papers

University of Iowa, Iowa City, IA

2011 to 2016

Writing Tutor, Writing Center

- Online tutoring (Fall 2013–Spring 2016)
 - Provided feedback for papers through online tutoring (10 hours per week)
- Online and Face-To-Face tutoring (Fall 2011, Spring 2012)
 - Met with 4 students for 4 hours of in-person tutoring sessions per week
 - Provided feedback for papers through online tutoring

Dear hiring committee,

I am writing to express my interest in the position of data science research consultant at the Data Science Consulting program, co-led by the Data Science and AI Academy and the Data & Visualization Services Department of the NC State University Libraries. I am currently a third-year PhD student at the Department of Statistics. Through my coursework, research, and other collaborative work, I have experience working with a wide variety of data science projects. Further, I have experience working at a university writing center for four years, which helped me realize my interest in consultation.

As a third-year PhD student in Statistics, I have worked in a wide variety of research projects. The research projects for my courses include tensor decomposition, using machine learning methods to predict critical temperatures for semiconductors, causal inference for multiple exposures and multiple outcomes, transfer learning, etc. My own research includes two very different projects. One project uses spectral methods to analyze the effects of multiple exposures with the application of analyzing health burdens of per- and polyfluoroalkyl substances (PFAS), and a second project designs a sequential sampling method to optimize how to best sample points for a surrogate model for computer experiments with the application of finding best conditions for stable and efficient flights. The data for the first project is geospatial in nature, while the data for the second project comes from a fluid dynamic simulator. I have also helped with a project on analyzing harmful algal bloom and with writing an online pamphlet for helping natural resource management specialists understand machine learning methods. I believe my experience in various data science projects and in collaborating with researchers outside statistics has prepared me for becoming a consultant.

Previously, I had been a writing tutor/consultant at the Writing Center at the University of Iowa for four years. The format of writing consultation is similar to the description of the current position. I met with students weekly to discuss the writing they brought in and helped them go through the revision process, and I also helped students asynchronously with the writing they submitted online. The consultation process was engaging and rewarding. I never knew what aspects of writing I would be discussing with the students on any given day, and the dynamic nature of consultation was deeply rewarding for me. I felt that I grew with the students in terms of writing and learned from their domain knowledge. Now that I am studying statistics, I am looking forward to gaining consultation experience in helping others working with data and thinking about what statistical tools are relevant and beneficial for the project on hand.

Besides my interest in consultation as described above, I hope to join a collaborative consulting team because I truly enjoy collaborating with researchers from other fields. I have worked with medical doctors, biological and agricultural engineers, and aerospace engineers. I enjoy the process of understanding what others are trying to achieve in their research projects, because I am interested in many fields outside of statistics. (What a great way to learn from experts!) Then I think of what data is needed and what statistical methods are appropriate and communicate my ideas. It is not always easy to explain statistical terms to people outside of statistics, but this exact challenge makes collaboration interesting. I learn to

explain the big picture and think of what visualization helps communicate results, which in turn helps me become a better communicator for my own research.

After earning my PhD degree, I hope to work as a researcher, whether in a university or a research institute. Even though I hope to lead my own methodological research, I am interested in collaborating with others. I believe the development of statistical methods should be motivated by real-world problems, and collaborating with others is crucial in understanding what problems need solved. Working as a data science research consultant would equip me with skills in consultation and collaboration, both of which will help me pursue my career as a researcher.

I hope I have clearly outlined my qualifications and deep interest in this position. Please feel free to reach out if you have any questions. I am looking forward to hearing from you.

Sincerely,

Shih-Ni Prim